The Poisson Process

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References

- S.M. Ross, “Stochastic Processes”: Chapter 2
- R. Wolff: Section 2.6
- M. Harchol-Balter: Chapter 11

Counting Process

- A stochastic process $\{N(t), t \geq 0\}$ is said to be a *counting process* if $N(t)$ represents total number of events that have occurred up to time t
- **Properties**:
  1) $N(t) \geq 0, \forall t$.
  2) $N(t)$ is integer-valued.
  3) If $s < t$, then $N(s) \leq N(t)$.
  4) For $s < t$, $N(t) - N(s)$ equals the number of events that have occurred in $(s, t]$.
- E.g.:
  - queuing theory (customers, calls),
  - reliability (failures, repairs),
  - finance (transactions),
  - physics (radioactive particle emissions),
  - neuroscience (spikes in neurons)

Counting Process (contd.)

- A counting process $\{N(t), t \geq 0\}$ is said to have *independent increments* if:
  - numbers of events that occur in disjoint time intervals are independent
- In particular:
  - for all $t, s \geq 0$, $N(t)$ and $N(t+s) - N(t)$ are independent
- A counting process $\{N(t), t \geq 0\}$ is said to have *stationary increments* if:
  - distribution of number of events that occur in any interval of time depends only on the length of the time interval
- That is:
  - for all $0 \leq t_1 < t_2$ and $s > 0$, $N(t_2 + s) - N(t_1 + s)$ has same distribution as $N(t_2) - N(t_1)$

Poisson Process

- The counting process $\{N(t), t \geq 0\}$ is said to be a *Poisson process* with rate $\lambda > 0$ if:
  i. $N(0) = 0$.
  ii. The process has independent increments.
  iii. The number of events in any interval of length t is a Poisson random variable with mean λt.
- Note that for all $s, t \geq 0$:
  - $P\{N(t+s) - N(s) = n\} = e^{-\lambda t} \frac{(\lambda t)^n}{n!}, n \in \{0,1,2, \dots\}$
- A Poisson process has stationary increments:
  - follows from iii.
- $E(N(t))$:
  - λt

Poisson Process (contd.)

- Recall:
  - The function f is said to be $o(h)$ if:
    - $\lim_{h \to 0} \frac{f(h)}{h} = 0$
- Following is alternative definition of Poisson process
- The counting process $\{N(t), t \geq 0\}$ is said to be a Poisson process with rate $\lambda > 0$ if:
  - *i.* $N(0) = 0$.
  - ii. The process has stationary and independent increments.
  - *iii.* $P\{N(h) = 1\} = \lambda h + o(h)$.
  - *iv.* $P\{N(h) \geq 2\} = o(h)$.
- Can be shown that this definition is equivalent to the one on previous slide:
  - proof omitted
- However, we will provide intuition for the fact that iii. and iv. imply that $N(t)$ is a Poisson r.v. with mean λt

Recall: Poisson Approximation for Binomial Random Variable

- **<u>Theorem:</u>** Let $n \to \infty$ and $p \to 0$ s.t. $np \to \lambda > 0$. Then:
  - ❑ $\lim \binom{n}{k} p^k (1-p)^{n-k} = \frac{e^{-\lambda}\lambda^k}{k!}$.
- **<u>E.g.:</u>**
  - ❑ $n = 10000$ telephone subscribers in a region
  - ❑ $p =$P(a subscriber initiates call in a given 10 s window)=10^{-4}
  - ❑ P($k = 10$ calls initiated in window):
    - ❑ $\approx \frac{e^{-1}}{10!}$

Poisson Process (contd.)

- Let $N(t)$ be a Poisson process
- Recall:
  - The process has stationary and independent increments.
  - $P\{N(h) = 1\} = \lambda h + o(h)$.
  - $P\{N(h) \geq 2\} = o(h)$.
- Subdivide the interval $[0, t]$ into k equal parts, where k is large
- $P\{2 \text{ or more events in at least one subinterval}\}$:
  - $\leq \sum_{i=1}^{k} P\{2 \text{ or more events in the } i\text{th subinterval}\}$
  - $= ko\left(\frac{t}{k}\right)$
  - $= t\left(\frac{o\left(\frac{t}{k}\right)}{\frac{t}{k}}\right) \to 0$ as $k \to \infty$
- Hence, $N(t)$ will equal number of subintervals in which an event occurs
- This number has binomial distribution with parameters k and $p = \frac{\lambda t}{k} + o\left(\frac{t}{k}\right)$
- Hence, by Poisson approximation for binomial r.v., $N(t)$ will be a Poisson r.v. with mean λt

Interarrival Time Distributions

- Let $\{N(t), t \geq 0\}$ be a Poisson process with rate λ
- Let:
  - X_1 denote time of first event
  - X_n denote time between $(n-1)$st and nth events, $n \geq 2$
- **Proposition**: X_n, $n \in \{1,2, \dots\}$ are i.i.d. exponential random variables with rate λ
- **Proof**:
  - $P(X_1 > t)$:
    - $P(N(t) = 0)$
    - $e^{-\lambda t}$
  - Hence, X_1 has exponential distribution with rate λ
  - $P(X_2 > t | X_1 = s)$:
    - $P(0 \text{ events in } (s, s+t] | X_1 = s)$
    - $P(0 \text{ events in } (s, s+t]\ )$ (by independent increments)
    - $e^{-\lambda t}$
  - Hence, X_2 has exponential distribution with rate λ and X_2 is independent of X_1
  - The result follows by repeating above argument

Waiting Time Distributions

- Let S_n be:
  - arrival time of nth event
  - also called waiting time until nth event
- Then S_n:
  - $\sum_{i=1}^{n} X_i$
- **Proposition**: PDF of S_n is:

  1) $f_{S_n}(t) = \lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!}$, for $t \geq 0$
- This is a *gamma distribution*
- **Proof***:*
  - One way: find moment generating function (MGF) of S_n and find its inverse using a table of MGFs
  - Another way is as follows
  - CDF of S_n is $P(S_n \leq t)$:
    - $P(N(t) \geq n)$
    - $\sum_{j=n}^{\infty} e^{-\lambda t} \frac{(\lambda t)^j}{j!}$
  - Differentiate w.r.t. t to get a telescoping sum, which simplifies to expression in 1)

Another Definition of Poisson Process

- Recall:
  - **Proposition**: $X_n, n \in \{1,2, \dots\}$ are i.i.d. exponential random variables with rate λ
- This gives us another way of defining Poisson process, which is as follows
- Consider a sequence $\{X_n, n \geq 1\}$ of i.i.d r.v.s, each being exponential with rate λ
- Consider a counting process $\{N(t), t \geq 0\}$, whose nth event occurs at time:
  - $S_n = \sum_{i=1}^{n} X_i$
- Then $\{N(t), t \geq 0\}$ is a Poisson process with rate λ
- $\{N(t), t \geq 0\}$ defined in this way has stationary and independent increments and $N(t)$ is a Poisson random variable with mean λt:
  - by memoryless property of exponential r.v. and facts on previous slide

Conditional Distribution of Arrival Times

- Consider a Poisson process $\{N(t), t \geq 0\}$ with rate λ
- Suppose we know that exactly one event has occurred in $[0, t]$
- Want distribution of its time of occurrence, X_1
- **Claim**: X_1 is uniformly distributed in $[0, t]$
- **Proof**:
  - For $s \leq t$, $P\{X_1 < s | N(t) = 1\}$:
    - $\frac{P\{1 \text{ event in } [0,s), 0 \text{ events in } [s,t)\}}{P\{N(t)=1\}}$
    - $\frac{P\{1 \text{ event in } [0,s)\} P\{0 \text{ events in } [s,t)\}}{P\{N(t)=1\}}$
    - $\frac{s}{t}$
- Next, we generalize above result

A Useful Result

- Let $Y_1, \dots, Y_n$ be i.i.d. continuous r.v.s with PDF $f(.)$
- $Y_{(k)}$: k'th smallest of $Y_1, \dots, Y_n$
- Recall: $Y_{(1)}, \dots, Y_{(n)}$ called order statistics of $Y_1, \dots, Y_n$
- Joint PDF of $Y_{(1)}, \dots, Y_{(n)}$ is $f(y_1, \dots, y_n)$:
  - $n! \prod_{i=1}^{n} f(y_i)$, for $y_1 \leq y_2 \leq \cdots \leq y_n$
- If $Y_1, \dots, Y_n$ are uniformly distributed in $(0, t)$, then joint PDF $f(y_1, \dots, y_n)$:
  - $\frac{n!}{t^n}$, for $0 < y_1 \leq y_2 \leq \cdots \leq y_n < t$

Conditional Distribution of Arrival Times (contd.)

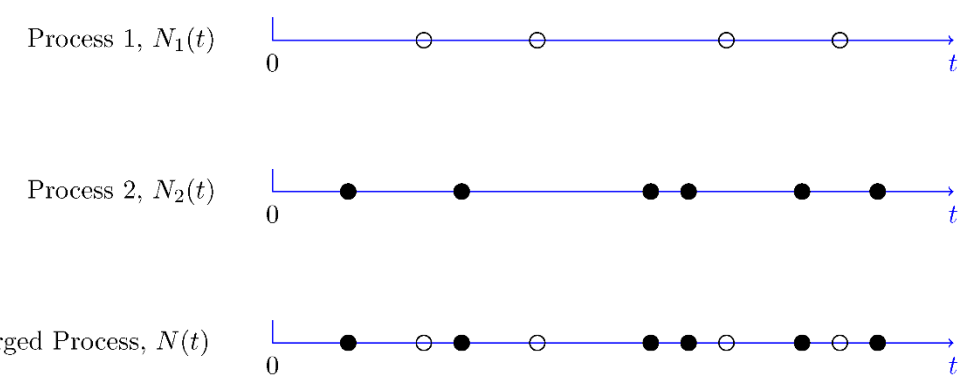
- Consider a Poisson process $\{N(t), t \geq 0\}$ with rate λ
- **Theorem**: Given that $N(t) = n$, the n arrival times $S_1, \dots, S_n$ have same distribution as order statistics corresponding to n independent r.v.s uniformly distributed in $(0, t)$
- That is:
  - given that n events have occurred in $(0, t)$, the times $S_1, \dots, S_n$ at which they occur, considered as unordered r.v.s, are distributed independently and uniformly in $(0, t)$
- **Proof**:
  - Let $0 < t_1 < t_2 < \cdots < t_{n+1} = t$
  - For $i \in \{1, \dots, n\}$, let h_i be small enough so that $t_i + h_i < t_{i+1}$
  - $P\{t_i \leq S_i \leq t_i + h_i, i \in \{1, \dots, n\} | N(t) = n\}$:
    - $\frac{P\{\text{exactly 1 event in } [t_i, t_i+h_i], i \in \{1,\dots,n\}, \text{no events elsewhere in } [0,t]\}}{P\{N(t)=n\}}$
    - $\frac{n!}{t^n} h_1 h_2 \dots h_n$
  - So, $\frac{P\{t_i \leq S_i \leq t_i + h_i, i \in \{1,\dots,n\} | N(t)=n\}}{h_1 h_2 \dots h_n} = \frac{n!}{t^n}$
  - Take limits as $h_i \to 0$, $i \in \{1, \dots, n\}$

Example

- Travelers arrive at a train depot in accordance with Poisson process $\{N(t), t \geq 0\}$ with rate λ
- The train departs at time t
- Want expected sum of waiting times of travelers arriving in $(0, t)$:
  - i.e., want $E\left[\sum_{i=1}^{N(t)}(t - S_i)\right]$
- $E\left[\sum_{i=1}^{N(t)}(t - S_i) \,|N(t) = n\right]$:
  - $nt - E\left[\sum_{i=1}^{N(t)} S_i \,|N(t) = n\right]$
- Let $U_1, \ldots, U_n$ denote a set of n independent uniform $(0, t)$ r.v.s
- $E\left[\sum_{i=1}^{N(t)} S_i \,|N(t) = n\right]$:
  - $E\left[\sum_{i=1}^{n} U_{(i)}\right]$
  - $E\left[\sum_{i=1}^{n} U_i\right]$
  - $\frac{nt}{2}$
- So, $E\left[\sum_{i=1}^{N(t)}(t - S_i) \,|N(t) = n\right]$:
  - $\frac{nt}{2}$
- $E\left[\sum_{i=1}^{N(t)}(t - S_i)\right]$:
  - $\frac{t}{2}E[N(t)]$
  - $\frac{\lambda t^2}{2}$

Merging Independent Poisson Processes

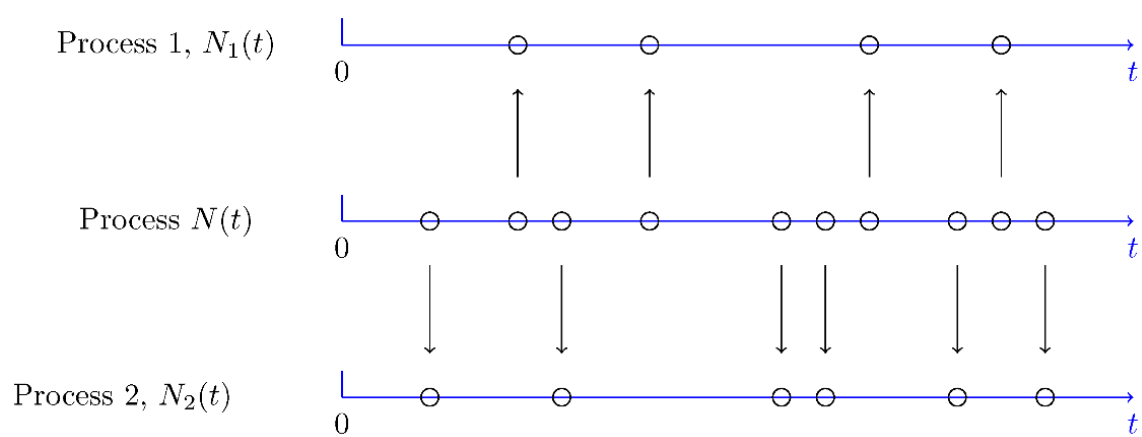
- **Theorem**:
  - Consider $k \geq 2$ independent Poisson processes, where process i has rate λ_i
  - The merge of the k processes is a Poisson process with rate $\sum_{i=1}^{k} \lambda_i$
- **Proof**:
  - For $i \in \{1, \dots, k\}$, let $N_i(t)$ be number of events in process i until time t
  - Let $N(t)$ be number of events in merged process until time t
  - For fixed t, $N_i(t)$ is Poisson$(\lambda_i t)$ for each $i \in \{1, \dots, k\}$
  - Hence, as proved earlier using characteristic functions, $N(t)$ is Poisson$\left(t \sum_{i=1}^{k} \lambda_i\right)$
  - Also, merged process has independent and stationary increments because each of the k original processes does



Ref:
https://www.probabilitycourse.com/chapter11/11_1_3_merging_and_splitting_poisson_processes.php

Poisson Splitting Property

- **Theorem**:
  - A Poisson process $\{N(t), t \geq 0\}$ with rate λ is split into $k \geq 2$ processes, $\{N_i(t), t \geq 0\}$, $i \in \{1, \dots, k\}$, as follows
  - Each arrival is assigned to process $i \in \{1, \dots, k\}$ w.p. p_i independently of the other arrivals, where $\sum_{i=1}^{k} p_i = 1$
  - Then for each $i \in \{1, \dots, k\}$, $\{N_i(t), t \geq 0\}$ is a Poisson process with rate λp_i and the k processes are independent



Ref: https://www.probabilitycourse.com/chapter11/11_1_3_merging_and_splitting_poisson_processes.php

Example 1

- An arrival stream of cars $\{N(t), t \geq 0\}$ is a Poisson process with rate λ
- W.p. p (respectively, $1 - p$), a car is a Ford (respectively, Chevrolet)
- Then arrival processes of Fords and Chevrolet are independent Poisson processes with rates λp and $\lambda(1 - p)$, respectively

Example 2

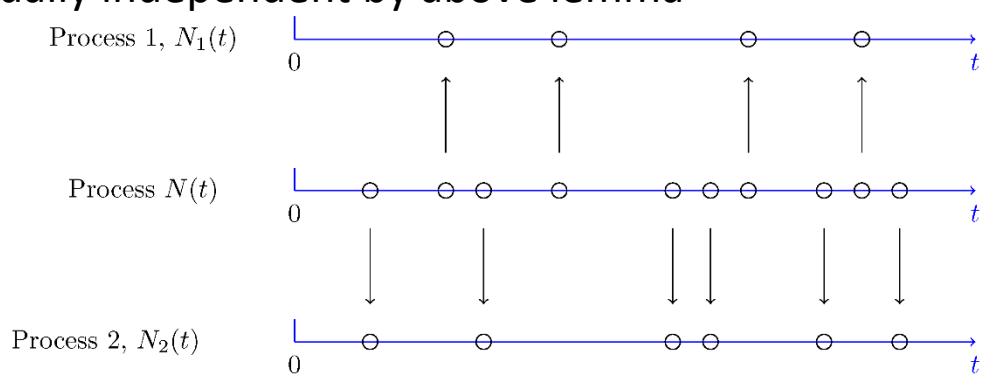
- Cars arrive at parking lot of concert hall according to Poisson process, $\{N(t), t \geq 0\}$, with rate λ
- With cars numbered in order of arrival, traffic cop at entrance directs:
  - odd-numbered cars north towards parking lot section 1
  - even-numbered cars south towards parking lot section 2
- Let $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ denote arrival streams of cars heading north and south, respectively
- From above theorem, does it follow that $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ are independent Poisson processes?
  - No, note that $N_2(t) = N_1(t)$ or $N_1(t) - 1$ for all t
- Why is above theorem not applicable?
  - Because the partitioning is not random

Poisson Splitting Property (contd.)

- **Lemma**:
  - Let N be a Poisson(λ) r.v.
  - We randomly partition N into $k \geq 2$ categories, where p_i is the probability of category $i \in \{1, \dots, k\}$ and $\sum_{i=1}^{k} p_i = 1$
  - Let N_i be the number in category i
  - Then $N_1, \dots, N_k$ are mutually independent r.v.s and N_i is Poisson(λp_i) for each $i \in \{1, \dots, k\}$
- **Proof**:
  - We prove the result for $k = 2$; general case left as an exercise
  - $P(N_1 = i, N_2 = j)$:
    - $P(N_1 = i, N_2 = j | N = i + j)\, P(N = i + j)$
    - $\binom{i+j}{i} p_1^i\, p_2^j \frac{e^{-\lambda}\lambda^{i+j}}{(i+j)!}$
    - $\left(\frac{e^{-\lambda p_1}(\lambda p_1)^i}{i!}\right)\left(\frac{e^{-\lambda p_2}(\lambda p_2)^j}{j!}\right)$
    - $P(N_1 = i)P(N_2 = j)$

Poisson Splitting Property (contd.)

- Recall: **Theorem**:
  - A Poisson process $\{N(t), t \geq 0\}$ with rate λ is split into $k \geq 2$ processes, $\{N_i(t), t \geq 0\}$, $i \in \{1, \dots, k\}$, as follows
  - Each arrival is assigned to process $i \in \{1, \dots, k\}$ w.p. p_i independently of the other arrivals, where $\sum_{i=1}^{k} p_i = 1$
  - Then for each $i \in \{1, \dots, k\}$, $\{N_i(t), t \geq 0\}$ is a Poisson process with rate λp_i and the k processes are independent
- **Proof**:
  - By above lemma, for every fixed $t \geq 0$, $N_i(t)$ is a Poisson$(\lambda t p_i)$ r.v.
  - We need to show:
    - *a)* $\{N_i(t), t \geq 0\}$ has stationary and independent increments, and
    - b) Increments of different processes $\{N_i(t), t \geq 0\}$, $i \in \{1, \dots, k\}$ are independent
  - a) follows from fact that $\{N(t), t \geq 0\}$ has stationary and independent increments
  - To show b), select subsets $V_i \subseteq [0, t]$, $i \in \{1, \dots, k\}$, and let N^{V_i} be number of events in V_i of the process $\{N_i(t), t \geq 0\}$
  - Partition $N(t)$ into the $2k$ categories V_i, $i \in \{1, \dots, k\}$ and $[0, t] \backslash V_i$, $i \in \{1, \dots, k\}$
  - The r.v.s N^{V_i}, $i \in \{1, \dots, k\}$, are mutually independent by above lemma



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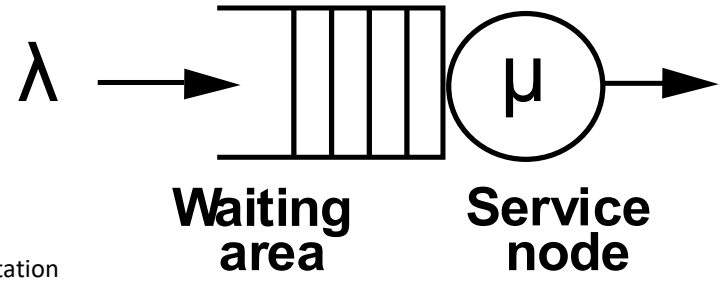
Kendall's Notation for Queues

- Standard notation used to describe queuing model
- Queueing model described as $A/S/c/K$, where:
  - A: inter-arrival time distribution,
  - S: service time distribution,
  - c: number of channels,
  - K: maximum number of customers in the *system* (waiting + being served)
- If fourth parameter not specified, then it is assumed that:
  - $K = \infty$
- Symbols in positions A and S denote specific families of distributions, e.g.:
  - M: Markovian or memoryless (Poisson process/ exponential service time)
  - D: deterministic or constant (fixed inter-arrival time/ fixed service time)
  - G: general distribution

Kendall's Notation for Queues (contd.)

- Recall:
  - Queueing model described as $A/S/c/K$, where:
    - A: inter-arrival time distribution,
    - S: service time distribution,
    - c: number of channels,
    - K: maximum number of customers in the system (waiting + being served)
- E.g.:
  - M/M/1:
    - Poisson arrival process, exponential service, single channel
  - M/G/∞:
    - Poisson arrival process, general service, infinite number of channels
  - G/G/1:
    - General arrival process, general service, single channel
  - M/G/3/10:
    - Poisson arrival process, general service, three channels, maximum of 10 customers allowed in the system



Ref: https://en.wikipedia.org/wiki/Kendall%27s_notation

Example: The M/G/∞ Queue

- Calls arrive at a telephone switchboard according to Poisson process with rate λ
- Call durations are i.i.d. with CDF $G(.)$ that has mean $\frac{1}{\mu}$, where $0 < \mu < \infty$
- Infinite number of phone lines available
- Let:
  - ❑ $N(t)$: number of calls that arrive during $[0, t]$
  - ❑ $M(t)$: number of calls in progress at time t
- Want to find distributions of $M(t)$ and $N(t) - M(t)$
- Given $N(t) = n$, arrival times of the n calls (considered as unordered r.v.s) are uniformly distributed and independent r.v.s
- Probability that a given call is in progress at time t, say p_t:
  - ❑ $\frac{1}{t}\int_0^t G^c(t-x)\,dx$
  - ❑ $\frac{1}{t}\int_0^t G^c(u)\,du$
- By Poisson splitting property:
  - ❑ $M(t)$ and $N(t) - M(t)$ are independent Poisson r.v.s with means $\lambda t p_t$ and $\lambda t(1 - p_t)$, respectively
- **Note**: We used the version of the Poisson splitting property for Poisson *r.v.s*, not Poisson *processes*

Example

- A device is subject to shocks that occur in accordance with Poisson process $\{N(t), t \geq 0\}$ with rate λ
- ith shock results in damage D_i, where:
  - $D_1, D_2, D_3, \ldots$ are i.i.d. with mean $E(D)$ and independent of the process $\{N(t), t \geq 0\}$
- Damage due to a shock decreases exponentially in time, i.e.:
  - If initial damage is D_i, then damage after time x is $D_i e^{-\alpha x}$
- Damages due to different shocks are additive, i.e., damage at time t is:
  - $D(t) = \sum_{i=1}^{N(t)} D_i e^{-\alpha(t-S_i)}$,
  - where S_i is arrival time of ith shock
- Want to find $E[D(t)]$
- $E[D(t)|N(t) = n]$:
  - $E\left[\sum_{i=1}^{n} D_i e^{-\alpha(t-S_i)} \,|N(t) = n\right]$
  - $E[D]e^{-\alpha t} E[\sum_{i=1}^{n} e^{\alpha S_i} \,|N(t) = n]$
  - $E[D]e^{-\alpha t} E[\sum_{i=1}^{n} e^{\alpha U_i} \,|N(t) = n]$,
  - $E[D]e^{-\alpha t} \frac{n}{t} \int_0^t e^{\alpha x}\, dx$
  - $E[D] \frac{n}{\alpha t}(1 - e^{-\alpha t})$
  - where $U_1, \ldots, U_n$ are i.i.d. uniform $[0, t]$ r.v.s
- So, $E[D(t)]$:
  - $\frac{\lambda E[D]}{\alpha}(1 - e^{-\alpha t})$

Reasons for Wide Use of Poisson Process in Practice

1) Markovian properties of Poisson process make it analytically tractable
2) In many cases, it is an excellent model. E.g.,
   a) In communications networks, e.g., in telephone system, it is a good model for sequence of times at which telephone calls are originated
      - although calls of a single user do not look like a Poisson process, aggregate over many users does
   b) Many physical phenomena behave in Poisson fashion, e.g., sequence of gamma ray emissions from radioactive substance

- Poisson process often appears in nature when we are observing aggregate effect of large number of individuals or particles operating independently
- Reason: *Limiting Theorem (*due to *Palm and Khinchin):*
  - ❑ Suppose we merge a large number, n, of i.i.d arrival processes, each with arrival rate $\frac{\lambda}{n}$, where each of the arrival processes is a renewal process with an arbitrary fixed interarrival distribution F
  - ❑ Then aggregate arrival process approaches Poisson process with rate λ
- Analogous to fact that many practical random processes are Gaussian due to Central Limit Theorem